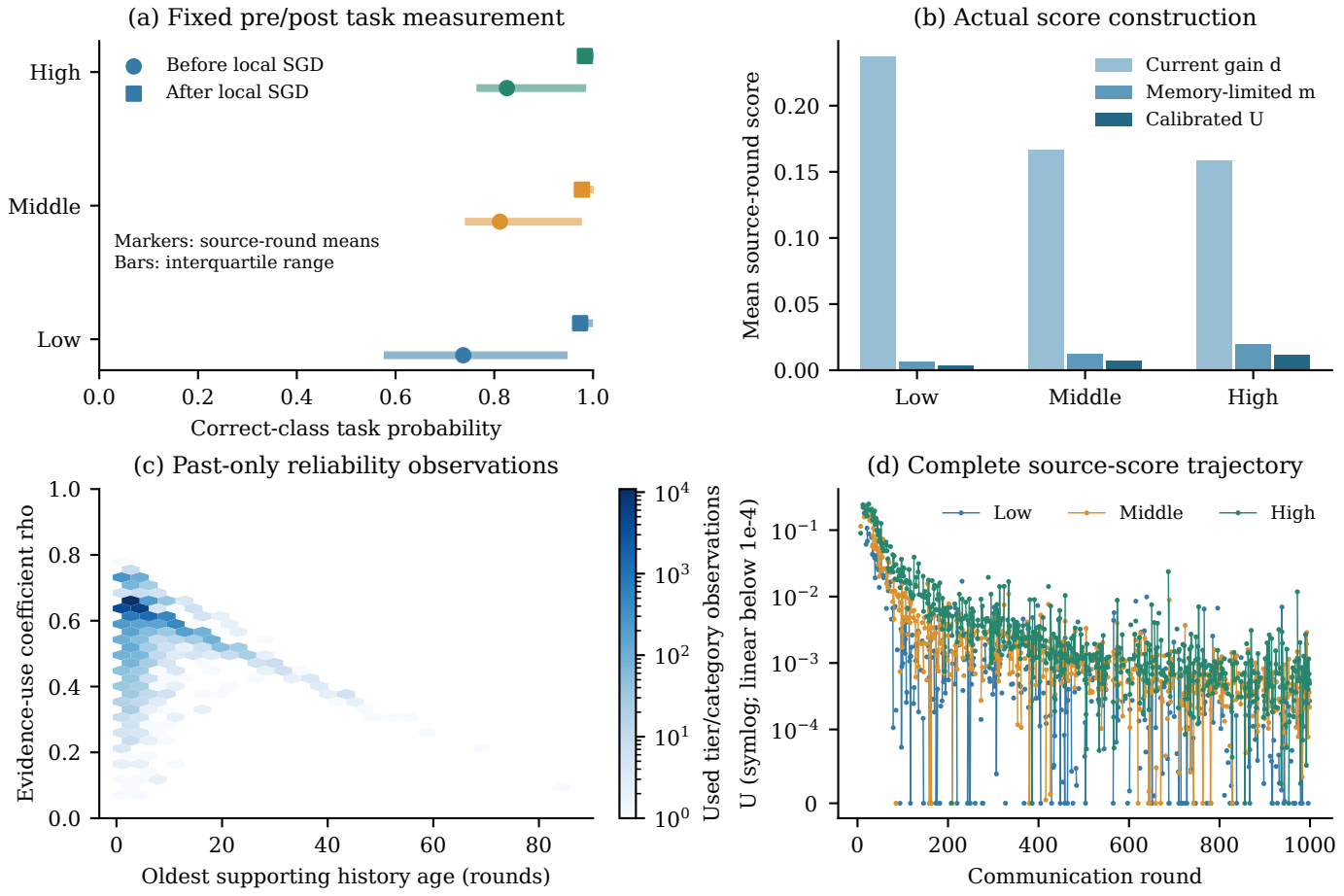

(a) Fixed pre/post task measurement
Before local SGD
After local SGD
High
Middle
Low
Markers: source-round means
Bars: interquartile range
0.0
0.2
0.4
0.6
0.8
1.0
Correct-class task probability
(b) Actual score construction
Current gain d
Memory-limited m
Calibrated U
Mean source-round score
0.00
0.05
0.10
0.15
0.20
Low
Middle
High
(c) Past-only reliability observations
Evidence-use coefficient rho
Oldest supporting history age (rounds)
Used tier/category observations
(d) Complete source-score trajectory
Low
Middle
High
U (symlog; linear below 1e-4)
Communication round